Measuring AI Value, Performance, and Operational Impact

OPS 310

AI tools should be evaluated not only by model accuracy, but also by whether they improve public health operations and outcomes. This module teaches learners to define measures for timeliness, quality, equity, staff burden, cost, user satisfaction, safety, trust, and sustainability. It helps agencies avoid adopting AI because it is novel and instead assess whether it creates measurable public value.

Level	intermediate/practitioner
Primary track	Operations and Data Quality Track
Appears in tracks	operations-data-quality

Learning Objectives

- Explain the purpose of measuring ai value, performance, and operational impact in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a AI value and impact measurement plan that can support readiness assessment, governance review, procurement, implementation, or monitoring.

Module Overview

AI tools should be evaluated not only by model accuracy, but also by whether they improve public health operations and outcomes. This module teaches learners to define measures for timeliness, quality, equity, staff burden, cost, user satisfaction, safety, trust, and sustainability. It helps agencies avoid adopting AI because it is novel and instead assess whether it creates measurable public value.

Why This Topic Matters for Public Health AI

Measuring AI Value, Performance, and Operational Impact matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

Core Concepts

AI value is the benefit created by an AI-supported workflow relative to its costs, risks, and alternatives. Value may include faster response, improved quality, reduced burden, better access, stronger equity, improved documentation, or better use of limited resources.

Model performance measures how well the AI system performs a technical task. Operational impact measures whether the workflow actually improves public health practice. A model can have strong technical metrics and still fail to reduce burden, improve timeliness, or support equitable outcomes.

Evaluation should be planned before implementation. Agencies should define baseline measures, target outcomes, data sources, review frequency, and decision rules for continuation, modification, scaling, or retirement.

Public Health Example

A health department may deploy AI to summarize case reports with the goal of reducing investigator burden. Accuracy alone is not enough to prove value. The agency should also measure review time, editing burden, error types, investigator satisfaction, case completion timeliness, escalation frequency, subgroup performance, and whether staff continue to trust and use the tool appropriately.

Technical Deep Dive

Value measurement begins with a theory of change. The team should explain how the AI capability is expected to improve the workflow and which outcomes should change. For example, a summarization tool may reduce review time, improve completeness, and allow staff to focus on complex cases. These assumptions should be tested rather than assumed.

Measures should include technical, operational, equity, and governance indicators. Technical indicators may include accuracy, error rates, latency, and uptime. Operational indicators may include timeliness, backlog, staff burden, user adoption, and quality. Equity indicators may include subgroup performance, access, language, disability, and community impact. Governance indicators may include documentation completeness, incident reports, and monitoring review completion.

Agencies should define action thresholds. If the tool produces too many errors, increases burden, performs poorly for a subgroup, or creates trust concerns, the agency should know whether to retrain, revise, restrict, pause, or retire the tool. Value measurement should support decisions, not merely reporting.

Risks, Failure Modes, and Guardrails

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

Application to AI-Supported Workflows

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

Reflection Questions

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

Practical Exercise

Create a AI value and impact measurement plan for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

Expected Artifact or Evidence

The expected artifact is a completed AI value and impact measurement plan. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Expected Assignment

- The expected artifact is a completed AI value and impact measurement plan. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Knowledge Check

- 1. What is the best reason to address measuring ai value, performance, and operational impact before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

References and Resources

- CDC Evaluation Framework: https://www.cdc.gov/evaluation/
- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST AI RMF Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- Agency performance management, quality improvement, equity, and evaluation policies